

Command words

What each word asks you to do, and what the examiners reward

Write full sentences and give each point its reason or detail. Answer the question on the paper, using its scenario.

Which words carry the marks

Every Paper 1 and Paper 3 question part from 2022 to March 2026: 925 parts in 56 papers. A part counts under the command word in its own wording.

Word	Cambridge's definition (syllabus p. 66)	Parts	Share of marks	
Describe	state the points of a topic/give characteristics and main features	409	38.4%	
Explain	set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence	176	17.5%	
Discuss	write about issue(s) or topic(s) in depth in a structured way	80	15.4%	
Evaluate	judge or calculate the quality, importance, amount, or value of something	25	4.8%	
Analyse	examine in detail to show meaning, identify elements and the relationship between them	19	3.4%	
Justify	support a case with evidence/argument	20	3.1%	
Complete	an instruction that is not in the syllabus command-word table	17	3.0%	
Identify	name/select/recognise	28	2.6%	
Compare	identify/comment on similarities and/or differences	16	2.3%	
Draw	an instruction that is not in the syllabus command-word table	9	1.2%	
Write	an instruction that is not in the syllabus command-word table	7	0.9%	
State	express in clear terms	12	0.8%	
Give	an instruction that is not in the syllabus command-word table	11	0.8%	
Contrast	identify/comment on differences	6	0.6%	

The shares above add up to 94.7%. The other 5.3% is 77 parts with no command word of their own (4.4%), such as '(b) Trojan' under an Identify question, and List, Suggest, Define and Name (0.9% together).

Answer in full sentences

1. No Cambridge document for 9626 tells you to answer in bullet points. The Cambridge guide for students tells you to answer in sentences.

"Answer the questions in sentences. Never give 'one word' or one 'phrase' answers"
(Cambridge Learner Guide for 9626, p. 15).

2. Examiner reports give this advice in almost every exam series. Use full sentences for Describe and Explain answers as well as for the long questions.

"Candidates should also be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences"
(Examiner report, November 2025, Paper 9626/32, p. 16). *"Candidates should write full sentences and avoid bulleted short statements in their responses"* (Examiner report, June 2022, Paper 9626/31, p. 13).

3. Bullets usually leave out what earns the marks. A bullet is often a bare point with no reason or detail. Most of the marks are for the reason or detail.

"the use of bullet points rarely produces little more than simple points or short statements with no explanations or descriptions" (Examiner report, June 2018, Paper 9626/32, p. 17). *"Short statements or bulleted responses that do not expand on facts cannot be awarded the higher marks because these are not discussions"*
(Examiner report, November 2022, Paper 9626/31, p. 14).

4. For Discuss, Evaluate and Analyse, the shape of the answer matters too. Bulleted lists and tables of statements cannot get most of the marks. Separate lists of benefits and drawbacks have the same problem when each item is one word or a very short sentence.

"learners who produced bulleted lists or tables of statements were unable to access most of the marks"
(Examiner report, March 2025, Paper 9626/32, p. 10). *"This comment also applies to answers that have separate lists of benefits and drawbacks, but the lists consist of single word or very brief sentences"*
(Examiner report, June 2025, Paper 9626/32, p. 22).

5. The mark schemes once printed this rule too. From 2021 to 2023, almost every Paper 1 and Paper 3 mark scheme limited bulleted answers to Discuss, Evaluate or Analyse questions. On an 8-mark question, the limit was 6 marks, however good the points were. Shorter questions had lower limits, for example 3 of 4 marks.

"Max. six marks if bullets/list of points Must have expansions or comparisons to be a proper discussion"
(Mark scheme 9626/12, March 2022, p. 4). *"three marks if bullets/list of points"*
(Mark scheme 9626/12, November 2022, p. 7).

6. In 2024, only the March and June Paper 3 mark schemes still printed the limit. From November 2024, the mark schemes leave it out. The 2025 examiner reports still say that bulleted lists and tables cannot get most of the marks.

"Max 6 marks if bullets/list of points" (Mark scheme 9626/31, June 2024, p. 8). *"learners who produced bulleted lists or tables of statements were unable to access most of the marks"*
(Examiner report, March 2025, Paper 9626/32, p. 10).

7. One Paper 1 question in November 2025 asked for the steps of planning something. The examiners accepted a list of steps, sometimes bulleted, or a description. Each step had to show a decision. Everywhere else, the examiner reports ask for full sentences.

"In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable"
(Examiner report, November 2025, Paper 9626/12, p. 7).

How the marks are given

Points and their detail

1. You do not lose marks for grammar or spelling. Examiners give marks for correct points and do not take marks away for errors. Your meaning must still be clear.
"marks are not deducted for errors" (Mark scheme 9626/12, June 2025, p. 2). "answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous" (Mark scheme 9626/12, June 2025, p. 2).
2. Short sentences are fine. Put one clear point in each sentence. Each point is then easy for the examiner to mark and easy for you to check.
"Keep most sentences short and focused. That will help with clarity and in making every point stand out for the examiner" (Cambridge Learner Guide for 9626, p. 15).
3. Some mark schemes give one mark for a point and a further mark only for its reason or detail. They label the first mark (1st), with the linked follow-on marks under it. Some mark schemes say that the follow-on marks need the first one. One examiner report says the further marks were 'dependant on this answer'. Cambridge's mark-scheme key does not define (1st).
"(only award if either of previous two marks given)" (Mark scheme 9626/12, November 2025, p. 6). "Two further marks were then dependant on this answer" (Examiner report, June 2024, Paper 9626/11, p. 4).
4. One mark scheme limits an answer made only of bare points. On that 3-mark question, answers that are all first marks get at most 2 marks.
"MAX two marks if answers are all only first marks" (Mark scheme 9626/13, November 2023, p. 4).

Answer the question on the paper

1. Answer the question that the paper asks, about the scenario it gives. The full range of marks is only for answers that use the scenario. Do not write everything you know about one word in the question.
"the full range of marks is only available to candidates for answers referring to the scenario in the questions" (Examiner report, November 2025, Paper 9626/32, p. 16). "Centres must ensure that candidates target their answers to the questions as they are written and not at 'key' words in the questions" (Examiner report, June 2024, Paper 9626/31, p. 17).
2. Do not write out mark scheme bullets that you learnt by heart. Examiners notice this. A list of learnt points does not answer a new question.
"Examiners saw many scripts where candidates listed sets of bullet points that appeared to have been learnt directly from previous mark schemes" (Examiner report, November 2019, Paper 9626/11, p. 1).
3. Use the number of marks as a clue to how much to write. It tells you how many separate points, or how much detail, the examiner expects.
"Look at how many marks are allocated to a question. This gives you a clue as to how much work is expected" (Cambridge Learner Guide for 9626, p. 15). "the number of marks available for each question is a useful indicator of how much detail or how many descriptions should be given in the answers" (Examiner report, March 2025, Paper 9626/32, p. 8).
4. Never leave a question blank. A few sentences on a topic you are unsure of can still earn marks.
"Writing a few sentences, even if little is known about the topic, may gain valuable marks towards the overall total"

How to read a mark scheme

From June 2024, Paper 3 mark schemes print a list that explains their signs. Paper 1 mark schemes use the same signs but do not print the list. Where the list explains a sign, the table uses its words. For the other signs, the meaning comes from how the mark schemes use them.

Sign	What it means	Example from a mark scheme
/	The words on each side are alternatives. Any one of them earns the mark.	"/ separates alternative words / phrases within a marking point" (Mark scheme 9626/31, June 2025, p. 6) "Real time processing produces an extremely/fast response" (Mark scheme 9626/11, November 2025, p. 8)
//	Two different answers that earn the same mark. Give either one.	"// separates alternative answers within a marking point" (Mark scheme 9626/31, June 2025, p. 6)
()	You do not need the words in brackets. They only show the context of the point.	"() the word / phrase in brackets is not required, but sets the context" (Mark scheme 9626/31, June 2025, p. 6)
Underlined word	You must use this word, or a form of it such as a plural.	"Underline actual word given must be used by candidate (grammatical variants accepted)" (Mark scheme 9626/31, June 2025, p. 6)
max, MAX	The most marks that a group of points can earn.	"max indicates the maximum number of marks that can be awarded" (Mark scheme 9626/31, June 2025, p. 6)
(1)	One mark for that point.	"One mark per bullet point to a maximum of five marks" (Mark scheme 9626/12, March 2025, p. 4)
Four from:, Six from:	Any of the listed points can earn marks, up to that number.	"Four marks from" (Mark scheme 9626/12, June 2025, p. 4)
(1st)	The first mark in a chain. The dashed lines under it are follow-on marks for adding detail to that point. The printed list does not explain it. Some mark schemes say a follow-on mark needs the first one.	"First mark – any reasonable reason why a checker may miss issues when comparing two documents (1st)" (Mark scheme 9626/12, March 2025, p. 4) "(only award if either of previous two marks given)" (Mark scheme 9626/12, November 2025, p. 6)
only award if, NOT awardable if	This mark depends on another mark. Without the first point, this one scores nothing.	"NOT awardable if BP4 or BP5 not awarded" (Mark scheme 9626/13, June 2025, p. 7)
e.g., Content areas could include	Examples only. Other correct answers also get marks.	"Content areas could include" (Mark scheme 9626/31, June 2025, p. 12) "credit is given for valid answers which go beyond the scope of the syllabus and mark scheme" (Mark scheme 9626/12, June 2025, p. 2)
accept, allow	This different wording also earns the mark.	"accept either 'finds' or 'stops new infections'" (Mark scheme 9626/12, March 2023, p. 5) "(allow concept)" (Mark scheme 9626/12, June 2025, p. 4)

do not accept	This wording earns nothing, even though it looks close.	"Sensor(s) are used to gather data (DO NOT ACCEPT 'COLLECT')" (Mark scheme 9626/11, June 2023, p. 7)
converse, award once only	The opposite statement counts on the other side, for example as a drawback instead of a benefit. It earns one mark only, not one on each side.	"(allow converse, but award once only)" (Mark scheme 9626/11, June 2024, p. 8)
Max N marks if bullets/list of points	A lower limit for answers written as a list. Mark schemes printed it from 2021 to 2024.	"Max. six marks if bullets/list of points" (Mark scheme 9626/12, March 2022, p. 4)
Note: brand names	Name the type of software or hardware, not the product.	"No marks are awarded for using brand names of software packages or hardware." (Mark scheme 9626/31, June 2025, p. 6)

The words at a glance

Command word	What to do
Describe	Say what it is, what it does or how it happens, with one separate point for each mark. Give reasons only when the question asks for them.
Explain	Make a point, then give its reason with a linking word such as 'because' or 'so'. Do this for each mark you need.
Discuss	Write paragraphs that give benefits and drawbacks of the thing in the question. Expand each point with its reason or effect.
Evaluate (and Assess)	Give points for and against, each with a consequence and how much it matters here. End with a reasoned overall judgement.
Analyse	Break the topic into parts. For each part, say what it leads to in the question's situation.
Justify	Argue only for the choice in the question. Give each benefit with the reason it helps in this scenario.
Compare and Contrast	For Compare, give similarities and differences. For Contrast, give differences only, and write every point about both things.
Identify, State, Give, Define, Name, Suggest	Give the exact term or fact, in the number of items the question asks for. Leave out explanations unless the question also says describe or explain.
Complete, Draw, Write (algorithms and flowcharts)	Use the syllabus keywords and flowchart symbols. Set counters before the loop, update them inside it, and close every structure.

Describe

Describe. Cambridge's definition: "state the points of a topic/give characteristics and main features"

Syllabus 2025 to 2027, p. 66

Check what you must describe

1. Describe asks what something is, what it does or how it happens. Explain adds why. You need a reason in a Describe answer only when the question asks for one, for example 'Describe, with reasons'. Then you cannot get full marks without the reasons.
"Explain set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence" (Syllabus 2025 to 2027, p. 66). "Must have at least TWO reasons to obtain full marks" (Mark scheme 9626/11, June 2023, p. 5).
2. Examiner reports often say that students mix up Describe and Explain. The mix-up costs marks in both directions. If the question asks why, a description of the thing itself gets few or no marks.
"Many candidates failed to differentiate between, for example, questions where they were asked to describe and questions where they were asked to explain" (Examiner report, June 2024, Paper 9626/11, p. 1). "where a question asks why a protocol or process would be used, there is usually little or no credit given for describing the protocol or process itself since this does not answer the question 'why?'" (Examiner report, June 2022, Paper 9626/31, p. 13).
3. For 'Describe how X is set up' or 'the process of', give the steps in order. Details of how X is used earn nothing there.
"Good answers detailed the steps that would be taken to set up an electronic wallet. No details of the use of the wallet were required to gain marks" (Examiner report, June 2022, Paper 9626/32, p. 16).
4. For 'Describe the features' or 'the functions', say what X is or does. For 'Describe the benefits', give benefits and leave out uses. An answer about the wrong one gets no marks, even if it is true.
"some focused on the use of the internet and remote learning, neither of which are features of a MOOC" (Examiner report, November 2022, Paper 9626/31, p. 14). "Common mistakes included descriptions of uses, inaccurate references to network performance and to illegal activities" (Examiner report, March 2025, Paper 9626/32, p. 9).

One mark for each point

1. In most Describe mark schemes, each correct point earns one mark. A 4-mark Describe question needs four separate points. One long sentence with one point in it is not enough.
"As a description is generally a one-mark answer, candidates had to give three separate descriptions for full marks to be awarded" (Examiner report, November 2025, Paper 9626/11, p. 1). "Question 1 has four marks available for describing the structure of a cellular network so at least four points need to be made in the answer for full marks" (Examiner report, November 2025, Paper 9626/33, p. 19).
2. Some mark schemes give one mark for a point, labelled (1st), and another for more detail about it, labelled (1). The (1st) label works as in the general rules at the front. On those questions, four short points can get 4 marks. Two points with more detail can also get 4 marks. 83 of the 388 Describe mark schemes from 2022 onward use (1st). None before 2022 use it.
"It is interrogated by the inference engine (1st) – Using relevant rules/rules that apply (1)" (Mark scheme 9626/11, November 2025, p. 5). "This meant that candidates could gain full credit by either giving four descriptions or expanding on fewer descriptions" (Examiner report, June 2025, Paper 9626/33, p. 26).
3. If the question asks for a set number of things, for example 'Describe two reasons', the examiner marks only that many answers. Extra answers do not help. If you join two ideas with 'and', the second idea can count as a separate answer. Give each point its own sentence.
"If, for example, the question asks for two reasons why an action may be taken, only the first two answers given by the candidate may be awarded marks" (Examiner report, November 2025, Paper 9626/11, p. 1). "Typically, where candidates

join two concepts together using 'and', the concept after the word 'and' is treated as a second answer, unless the piece after the 'and' is clearly an explanation or clarification of the piece before the 'and'"

(Examiner report, November 2025, Paper 9626/11, p. 1).

Say what each thing is or does

1. Say what each thing is or does. Examiner reports say that answers which only name components do not get the higher marks. One mark scheme allows a maximum of 3 of 5 marks for naming events without descriptions.

"Max three for naming valid events without descriptions" (Mark scheme 9626/32, June 2024, p. 6). "Many responses indicated that candidates knew the names of the components but were unable to describe them in the detail required to gain the higher marks" (Examiner report, June 2022, Paper 9626/13, p. 9).

2. Do not repeat the question. Answers such as 'White box testing is used to test the code' or 'a sound sensor measures sound' get no mark. Add a detail that the question does not give you.

"Further, many candidates repeated the question, writing that 'white box testing is used to test the code'. In general, statements that repeat the question are not given credit" (Examiner report, June 2024, Paper 9626/32, p. 22).

"Candidates should avoid simply repeating the question in their response without further expansion; for example, a humidity sensor measures humidity and a sound sensor measures sound"

(Examiner report, June 2022, Paper 9626/12, p. 5).

3. Vague words such as 'more expensive' get no marks on their own. Name the exact cost, or say exactly what is cheaper.

"Some weaker responses contained responses such as 'more expensive' which is not enough at this level and statements such as this should always be expanded" (Examiner report, June 2022, Paper 9626/13, p. 8). "However, where candidates stated that a specific, named, cost could be avoided, this was awarded"

(Examiner report, June 2025, Paper 9626/12, p. 6).

4. Use the correct technical term. Examiners say that slang often makes an answer meaningless.

"The use of what may be considered slang often renders answers meaningless"

(Examiner report, March 2025, Paper 9626/12, p. 1).

5. Write descriptions as full sentences, with one point in each sentence. The Paper 3 reports say that a description needs full sentences. On one Paper 1 question, the examiners accepted a list of steps for a 'describe the process' question. Use full sentences on both papers.

"Descriptions or explanations can only be conveyed in full sentences"

(Examiner report, June 2022, Paper 9626/31, p. 13). *"In the first, candidates listed (sometimes by creating a bulleted list) steps that would be taken, whilst others described the steps that had been taken. Either method was acceptable"*

(Examiner report, November 2025, Paper 9626/12, p. 7).

Fit the scenario

1. Fit your points to the scenario. The question may be about a nuclear power plant or a bank's new customers. A point that is true in general but wrong for the scenario loses the mark. A prepared answer that ignores the question gets only low marks.

"context also provides a restriction, as certain answers may be acceptable for one scenario, but not for another"

(Examiner report, November 2025, Paper 9626/11, p. 1). *"those who provided an answer they had prepared for, rather*

Worked example

Q4(b) [4 marks] "(b) Describe how a knowledge base is used in an expert system."

Question paper 9626/11, November 2025, p. 4

Part (a) asked what is meant by the term knowledge base.

Full marks

The knowledge base holds the facts about the subject area of the expert system.¹ A human expert in that subject supplied these facts.² When a user enters a query, the inference engine interrogates the knowledge base.³ The inference engine only uses the rules that apply to that query.⁴

Where the marks come from ¹"Knowledge base holds facts (1)". ²"gathered from an expert (1)". ³"It is interrogated by the inference engine (1st)". ⁴"Using relevant rules/rules that apply (1)". (Mark scheme 9626/11, November 2025, p. 5)

Weak answer, about 1 or 2 of 4

The knowledge base stores information and rules. The expert system uses it to make decisions.

'Stores rules' matches a mark scheme point. 'Stores information' may get the 'holds facts' mark. It misses three points: the facts come from an expert, and the inference engine interrogates the knowledge base with only the relevant rules.

"For example, whilst many candidates stated that the knowledge base holds facts, very few stated that these were sourced from experts. Similarly, candidates were aware that rules are used to interrogate the knowledge base, but very few stated that only the relevant rules were used" (Examiner report, November 2025, Paper 9626/11, p. 2).

Worked example

Q4(a) [4 marks] "The directors of a company wish to create a database of customers. They are not sure whether they should ask a computer programmer to produce custom written software or they should purchase off-the-shelf software. (a) Describe the features of custom written software."

Question paper 9626/13, June 2022, p. 4

Full marks

A programmer develops custom written software specially for one specific company, here the directors' company.¹ The programmer makes it to fit that company's particular preferences and needs, such as the exact customer details the directors want to store.² The company gets support directly from the programmers who created it.³ The company that commissions the software has to pay the development costs.⁴

Where the marks come from ¹"Software that is specially developed for a specific company/business". ²"It is made to accommodate that customer's particular preferences and needs". ³"Support is obtained directly from the creators".

⁴"Development costs have to be paid for by client for whom its written/ commissioning company/business".

(Mark scheme 9626/13, June 2022, p. 4)

Weak answer, about 0 of 4

Custom written software is more expensive. It is better than off-the-shelf software.

'More expensive' is a vague comparison with no detail. 'Better' is an opinion and names no feature. The cost point earns a mark only when it says who pays the development costs.

"Some weaker responses contained responses such as 'more expensive' which is not enough at this level and statements such as this should always be expanded" (Examiner report, June 2022, Paper 9626/13, p. 8).

Explain

Explain. Cambridge's definition: "set out purposes or reasons/make the relationships between things clear/say why and/or how and support with relevant evidence" Syllabus 2025 to 2027, p. 66

Statement, then reason

1. An explanation has two parts. First, you make a statement. Then you give a reason, a purpose or more detail about that statement.

"An explanation requires a statement or description followed by a reason or purpose or further expansion of that statement" (Examiner report, June 2024, Paper 9626/32, p. 23). *"An explanation is statement of fact with a reason or expansion giving its purpose"* (Examiner report, March 2024, Paper 9626/32, p. 8).

2. Join the statement to its reason with a linking word such as 'so', 'so that', 'because' or 'which means'. The mark schemes and examiner reports put these words where the reason starts. Write the linking word in your answer too.

"The lower the sample rate, the smaller the file size (1st) – because there is less data stored (1)" (Mark scheme 9626/12, March 2024, p. 9). *"stating the task 'to check that the project has ended' should have been followed by a reason such as 'so that all stakeholders and clients can agree that no further additions or changes are necessary'"* (Examiner report, June 2025, Paper 9626/33, p. 25).

3. Write explanations in full sentences. A sentence keeps the point and its reason together.

"Candidates should be encouraged to write full sentences and discouraged from writing bulleted, short statements in their responses. Descriptions or explanations can only be conveyed in full sentences" (Examiner report, November 2022, Paper 9626/31, p. 12).

First marks and follow-on marks

1. Some Explain mark schemes label a statement (1st) and the reason or detail under it (1). The (1) mark is only for the reason or detail you add to that statement. The (1st) label works as in the general rules at the front. 176 Explain parts appeared from 2022 to March 2026, and 174 have a mark scheme. 38 of those 174 mark schemes use (1st).

"The sampling rate can affect the quality of audio files/how the audio sounds (1st) – because it captures sounds at discrete points (and not the whole sound) (1)" (Mark scheme 9626/12, March 2024, p. 9). *"Access control can be based on attribute of user within company (1st) – so that they can only access data appropriate for their job/task/role (1)"* (Mark scheme 9626/32, March 2024, p. 7).

2. A statement with no reason can still get its one mark, and no more. If every point you write is only a statement, the mark scheme can stop you getting full marks.

"A less detailed answer, such as 'they can learn at their own pace' can be given credit as it is a correct statement of the effect, but it lacks the detail to gain any further credit because it does not add an expansion" (Examiner report, June 2024, Paper 9626/32, p. 23). *"MAX two marks if answers are all only first marks"*

3. Sometimes the mark scheme links a reason or detail to a first point. The reason then earns its mark only if you also make that first point. In one encryption question, the two further marks went only to answers that first said encryption scrambles the data. In a 2-mark question, one full point and its reason can be the whole answer.

"A relatively easy first mark was given for any answer that stated that the data is scrambled. Many candidates achieved this mark. Two further marks were then dependant on this answer" (Examiner report, June 2024, Paper 9626/11, p. 4).

"Max two for one complete explanation" (Mark scheme 9626/12, March 2024, p. 9).

Reasons that fit the question

1. Explain asks why or how. Examiner reports say that many answers described what something is and lost marks.

"Most responses gave descriptive answers rather than explanations"

(Examiner report, November 2023, Paper 9626/12, p. 6). *"there is usually little or no credit given for describing what the point or process is since this does not answer the question 'how'"*

(Examiner report, November 2022, Paper 9626/31, p. 12).

2. A common mistake is to leave out the reason. A short phrase such as 'makes the code more readable' does not get full marks.

"A common mistake was to omit a reason or expansion to the response – the question asked for explanations"

(Examiner report, November 2023, Paper 9626/32, p. 16). *"a significant number only stated that comment make the code more readable so did not gain the full marks"* (Examiner report, March 2024, Paper 9626/32, p. 8).

3. Use exact technical words. In one encryption question, many answers said encrypted data 'cannot be understood'. A similar number said it 'cannot be read'. Only 'cannot be understood' got the mark.

"Many candidates made the point that encrypted data can not be understood, but an approximately equal number stated that encrypted data could not be read, and were not awarded marks"

(Examiner report, June 2024, Paper 9626/11, p. 4).

4. If the question has a scenario, use it. Give reasons that fit the situation in the question. General facts alone are not enough.

"As with all scenario-based questions, candidates are advised to consider the significance of this scenario"

(Examiner report, November 2023, Paper 9626/13, p. 9). *"The full range of marks is only available to candidates for answers referring to the scenario in the questions"* (Examiner report, November 2022, Paper 9626/31, p. 12).

Worked example

Q3(c) [3 marks] "Mainframe computers were developed from the need for computers to process data in tasks such as carrying out a census. (c) Explain why mainframe computers are used to process the data from a census." Question paper 9626/13, November 2023, p. 3

Full marks

A census collects a very large amount of data that the computer has to process,¹ as countries tend to have a lot of people and each person adds more data.² The computer also needs to process all of this data in a short time, so a mainframe with a high processing speed does the work.³

Where the marks come from ¹ "The amount of data that has to be processed is very large (1st)". ² "as countries tend to have a lot of people (1)". ³ "All data needs to be processed in a short period of time (1st)".

(Mark scheme 9626/13, November 2023, p. 4)

Weak answer, about 2 of 3

There is a lot of data in a census. Mainframes are very reliable.

Both sentences are only first marks, with no reason after them. The mark scheme gives a maximum of 2 marks when all the answers are only first marks. The examiners also said that most answers stopped at 'a large amount of data'.

"Typically, however, responses tended to focus on that fact that there is usually a large amount of data"
(Examiner report, November 2023, Paper 9626/13, p. 9).

Worked example

Q10(a) [3 marks] "Explain how each of these methods keeps data safe from unauthorised users. (a) encryption" Question paper 9626/11, June 2024, p. 11

Full marks

Encryption scrambles the data,¹ so that anyone who intercepts it cannot understand it,² which means the data is of no use to a hacker or any other third party.³

Where the marks come from ¹"Data is scrambled (1st)". ²"so it is not understandable (1)". ³"of no use to a hacker/third party (1)". (Mark scheme 9626/11, June 2024, p. 9)

Weak answer, about 0 of 3

Encryption makes the data unreadable so hackers cannot read it.

It never says that encryption scrambles the data. That first mark is missing, and the two marks that depend on it go with it. The examiners wanted 'cannot be understood', and 'cannot be read' is the wrong idea.

"Rather than making data unreadable, encryption puts data into a state whereby it can not be understood"
(Examiner report, June 2024, Paper 9626/11, p. 4).

Discuss

Discuss. Cambridge's definition: "write about issue(s) or topic(s) in depth in a structured way"

Syllabus 2025 to 2027, p. 66

Open briefly, then expand each point

1. Some mark schemes give marks for a short definition or description of the thing in the question. 18 of the 77 Discuss mark schemes do this. Most allow one mark. June 2025 Paper 31 Q7 allows two. Write one clear opening sentence and move on.

"Max one for: • Definition of computer simulation (1)" (Mark scheme 9626/11, November 2025, p. 9). *"Max TWO marks:"*
(Mark scheme 9626/31, June 2025, p. 12).

2. Marks come from a list of points in the mark scheme. Many points earn a second mark when you expand them with a reason or an effect. The mark scheme shows the point as (1st) and the expansion as (1).

"Direct data source/direct data is reliable/trusted (1st) – because we collected it (1)"

(Mark scheme 9626/12, June 2024, p. 7). *"Must have expansions or comparisons to be a proper discussion"*

(Mark scheme 9626/12, March 2022, p. 4).

3. Do not make a point and stop. The March 2022 examiner report says most answers made valid points, but few expanded them. Two November 2022 mark schemes cap answers with no expansion at 6 of 8.

"Most responses gained credit for making valid points, but few went on to discuss the points they made by expanding on them" (Examiner report, March 2022, Paper 9626/12, p. 2). *"Max six marks if no attempt at expansion"*

(Mark scheme 9626/12, November 2022, p. 10).

4. No Discuss mark scheme from 2022 to 2025 gives a mark for a conclusion or for your opinion. The syllabus meaning of Discuss does not ask for one. The November 2019 examiner report used older wording. It said Discuss often needs a conclusion. A one-line conclusion does no harm. Spend your time on expanded points.

"Discuss write about issue(s) or topic(s) in depth in a structured way" (Syllabus 2025 to 2027, p. 66). *"Discuss questions require candidates to give the important arguments for and against, and often require a conclusion"*

(Examiner report, November 2019, Paper 9626/12, p. 4).

Cover both sides

1. Each side often has its own maximum. 38 of the 77 Discuss mark schemes print a MAX for one side or for each side. On 8-mark questions it is usually 6, sometimes 5 and once 7. On those questions, an answer with only benefits cannot get full marks.

"Benefits max 6" (Mark scheme 9626/12, November 2024, p. 9). *"Disadvantages MAX five from"*

(Mark scheme 9626/12, June 2023, p. 7).

2. 23 mark schemes, all from 2022 and 2023, need both sides for full marks. 20 of them ask for at least two points on each side. The other 3 are 4-mark and 6-mark questions, and one of each is enough. 4 more mark schemes from November 2023 cap an answer with only benefits or only drawbacks. The June 2025 examiner report says the same.

"Must have at least two of each to gain full marks" (Mark scheme 9626/12, March 2022, p. 4). *"answers that address only benefits or only drawbacks cannot achieve full marks"* (Examiner report, June 2025, Paper 9626/32, p. 22).

3. Do not write the same point twice, once as a benefit and again reversed as a drawback. The examiner gives the mark only once, and you lose time.

"repeated the list, but in reverse structure, so the positive aspects of one became the negative points of the other"

(Examiner report, June 2024, Paper 9626/12, p. 7). *"(Converse answer may also be given as an advantage – Award ONCE only)"* (Mark scheme 9626/12, June 2023, p. 7).

Write in paragraphs

1. Bullet points lose marks. Every 2022 and 2023 Discuss mark scheme caps a bulleted list. So do 6 of the 17 from 2024. That makes 48 of the 77. The cap is 6 of 8, 4 or 5 of 6, or 3 of 4.

"Max. six marks if bullets/list of points" (Mark scheme 9626/12, March 2022, p. 4). *"Max. three marks if bullets/list of points"* (Mark scheme 9626/12, November 2022, p. 7).

2. The 2025 mark schemes do not print the bullet cap. The 2025 examiner reports still say that bulleted lists and tables of statements cannot reach most of the marks. Two separate lists of single words or very short sentences have the same problem.

"learners who produced bulleted lists or tables of statements were unable to access most of the marks"

(Examiner report, March 2025, Paper 9626/32, p. 10). *"answers that have separate lists of benefits and drawbacks, but the lists consist of single word or very brief sentences"* (Examiner report, June 2025, Paper 9626/32, p. 22).

3. Write one paragraph of benefits and one of drawbacks. Put the reason or effect of each point in the same

sentence as the point.

"Candidates should be encouraged to write answers in full sentences"

(Examiner report, June 2025, Paper 9626/31, p. 17). *"The strongest answers were written in full prose style and included points which were expanded upon"* (Examiner report, March 2022, Paper 9626/12, p. 2).

Name the cost and the people

1. Say what the money pays for. The examiners treat 'expensive' or 'costs more' as too vague. Name the spending, for example travel to interviews or printing questionnaires.

"Where candidates stated that collecting direct data was expensive, this was considered too vague, but if candidates identified an area on which money would have to be spent, which would not otherwise be spent, this was awarded"

(Examiner report, June 2024, Paper 9626/12, p. 7). *"Many responses mentioned costs without going into sufficient detail to merit credit"* (Examiner report, March 2022, Paper 9626/12, p. 2).

2. Link each benefit and drawback to the people or organisation in the question. Answers that described types of technology or the stages of a process got little or no credit. So did answers about the wrong topic.

"Descriptions of different types of TEL, statements about benefits or drawbacks gained little or no credit without further comment on how these can affect the achievement of students" (Examiner report, March 2025, Paper 9626/32, p. 10).

"describing the stages of the data mining process or confusing data mining with another topic altogether"

(Examiner report, June 2025, Paper 9626/31, p. 19).

Worked example

Q7 [8 marks] "Data can be collected by using a direct or an indirect data source. Discuss the advantages and disadvantages of using direct data compared to indirect data."

Question paper 9626/12, June 2024, p. 8

Full marks

The person who will use direct data collects it first-hand (a primary source). Someone else collected indirect data for another purpose (a secondary source).¹ One advantage of direct data is that it is more reliable and trusted, because the user collected it and knows exactly how.² Also, the user collects only specific, relevant data, so no time goes on sorting through data they do not need.³ Direct data is also more up to date than indirect data, which someone may have collected years ago.⁴ However, a disadvantage is that the sample size may be small, whereas indirect data sources tend to be much larger sets of data.⁵ Collecting direct data also costs money that would not otherwise be spent, such as travel to interviews and printing questionnaires.⁶

Where the marks come from ¹ "Simple statement of the difference between the two (1): e.g". ² "Direct data source/direct data is reliable/trusted (1st) – because we collected it (1)". ³ "ONLY Specific/relevant data is collected (1)".

⁴ "Direct data is (more) up-to-date (1) Disadvantages MAX SIX". ⁵ "The sample size may be small (1st) – whereas indirect data sources tend to be larger sets of data (1)". ⁶ "Collecting direct data involves the expense of travelling/photocopying/printing questionnaires//any expense to do with a direct data method (1)".

(Mark scheme 9626/12, June 2024, p. 7)

Weak answer, about 2 of 8

Advantages: direct data is cheaper, relevant, up to date. Disadvantages: indirect data is not relevant and old, direct data is expensive.

It is a list of single words. 'Expensive' does not say what the money pays for. The disadvantages repeat the advantages in reverse, so they earn nothing new.

"Some candidates effectively wrote what was good about direct data compared to indirect data and then repeated the list, but in reverse structure, so the positive aspects of one became the negative points of the other"

(Examiner report, June 2024, Paper 9626/12, p. 7).

The examiner report says this question was about direct data, with indirect data only as a reference point.

Worked example

Q7 [8 marks] "Discuss the benefits and drawbacks of using data mining in businesses."

Question paper 9626/31, June 2025, p. 8

Full marks

Data mining is the use of software tools to analyse very large data sets.¹ Its aim is to discover trends, patterns and relationships in the data that people would not spot by hand.² Data mining can improve sales and profits, because the business can target its advertising at the types of customer most likely to buy.³ The information it extracts also gives managers a sound basis for their decisions, instead of guesswork.⁴ On the other hand, the extracted information may not be accurate, because it depends on how good the source data set is.⁵ The business must also protect the privacy of the individuals in the data and follow data protection laws, which adds work and risk.⁶

Where the marks come from ¹ "data mining (is a set of techniques and tools) used to from work on / analyse large data sets". ² "to discover trends / patterns / to find relationships". ³ "used to improve sales / profits (1st) by targeting customers / demographics (1)". ⁴ "provides a sound basis for decision-making". ⁵ "extracted information may not be accurate (1st) depends on source of data set (1)". ⁶ "must ensure privacy of individuals / conform to data protection regulations". (Mark scheme 9626/31, June 2025, p. 12)

Weak answer, about 2 or 3 of 8

A two-column table: Benefits | Drawbacks; trends | expensive; profit | privacy; decisions | storage.

Single words in a table name points but give no reasons or effects. There is no discussion and no expansion mark.

"Furthermore, if these answers are shown in a sketched table, they do not properly answer the question as there is no discussion" (Examiner report, June 2025, Paper 9626/31, p. 17).

Evaluate (and Assess)

Evaluate. Cambridge's definition: "judge or calculate the quality, importance, amount, or value of something"

Syllabus 2025 to 2027, p. 66

Assess. Cambridge's definition: "make an informed judgement" Syllabus 2025 to 2027, p. 66

Judge each point

1. Evaluate asks you to judge the importance or value of something. Discuss asks you to write in depth in a structured way. You can judge each point, or give one judgement on the whole answer.

"Discuss write about issue(s) or topic(s) in depth in a structured way Evaluate judge or calculate the quality, importance, amount, or value of something" (Syllabus 2025 to 2027, p. 66). "the judgement can be for individual points or for an overall judgement on the whole content of the answer" (Examiner report, November 2024, Paper 9626/33, p. 21).

2. The most common mistake is to describe how the thing works and never judge it. For each feature, say whether it helps or causes problems.

"Candidates seemed to have some understanding of each process, but preferred to describe each of these than actually evaluate their efficacy" (Examiner report, March 2023, Paper 9626/12, p. 3). "candidates were expected to make factual statements about project management software (PMS) and then expand on these with comments about whether these facts helped or hindered the project" (Examiner report, November 2023, Paper 9626/33, p. 21).

3. Good answers take each point through three steps: a fact, its consequence, and a judgement of how much it matters. Some mark schemes give a first mark for the point and more for the expansion. June 2025 Paper 13 Q11 allows at most 4 marks for first marks alone.

"Good answers gave a description of how compression was used or of the effects of the compression and then added a consequence followed by a judgment" (Examiner report, June 2025, Paper 9626/33, p. 26). "MAX four for 1st marks only" (Mark scheme 9626/13, June 2025, p. 9).

4. Some mark schemes give one mark for a short definition or description of the thing you evaluate. Write one sentence, then start judging.

"One mark for a description of encryption (e.g. data scrambled/converted to cyphertext)" (Mark scheme 9626/12, March 2024, p. 9). "Valid definition / description of video-conferencing in education" (Mark scheme 9626/31, June 2025, p. 15).

Points for and against

1. Write about both sides. 3 of the 24 Evaluate mark schemes, all from November 2023, cap an answer that is all for or all against at 6 of 8. 2 others need at least two points on each side for full marks.

"Max 6 if all for or all against" (Mark scheme 9626/32, November 2023, p. 10). "Must be at least two of each for full marks" (Mark scheme 9626/32, March 2023, p. 7).

2. One-sided answers lose marks. The two sides do not need to be equal. In June 2025, the examiners accepted more points for than against.

"Candidates tended to concentrate on the positive implications, rather than both the positive and the negative, and this did restrict the overall marks" (Examiner report, June 2025, Paper 9626/12, p. 8). "Typically, candidates gave more answers in support than against, but this was acceptable" (Examiner report, June 2025, Paper 9626/13, p. 12).

End with a judgement

1. 8 of the 24 Evaluate mark schemes give one mark for a conclusion. All 8 are from November 2024 or later. That is 8 of the 9 mark schemes since that date. End with a short overall judgement that brings your points together.

"One mark is available for a valid conclusion / judgement (must have considered both pros and cons)"

(Mark scheme 9626/32, November 2024, p. 15). *"Such discussions should end with a conclusion that revisits and brings together the points made"* (Examiner report, November 2024, Paper 9626/12, p. 4).

2. Leaving out the conclusion costs marks. The March 2024 examiner report (Paper 12, Q9) says students did not reach a realistic conclusion. This reduced their marks.

"candidates largely failed to come to any realistic conclusion about encryption, which reduced the marks scored on the question" (Examiner report, March 2024, Paper 9626/12, p. 3).

Paragraphs on the thing named

1. Write in full sentences and paragraphs. 9 of the 24 Evaluate mark schemes, all from 2022 to 2024, cap a bulleted list. The cap is usually 6 of 8, or 5 of 6 on 6-mark questions.

"A detailed answer is needed, which cannot be achieved by writing bullet points"

(Examiner report, March 2023, Paper 9626/12, p. 1). *"Max 6 marks if bullets/list of points"*

(Mark scheme 9626/32, March 2022, p. 6).

2. Judge the thing in the scenario the question gives. Answers about spreadsheets that never mentioned financial modelling did badly.

"To gain full credit, candidates must produce a reasoned evaluation that weighs up the advantages and disadvantages for the given scenario/context" (Examiner report, March 2023, Paper 9626/12, p. 1). *"candidates who failed to make any mention of financial modelling struggled to score well on this question"*

(Examiner report, June 2025, Paper 9626/13, p. 12).

3. Evaluate only what the question names. Compare it with other methods only if the question asks you to.

"It was not necessary to compare pilot with other possible methods of implementation because the question asked candidates to 'evaluate pilot implementation' and made no mention of the other methods"

(Examiner report, November 2023, Paper 9626/32, p. 17).

Worked example

Q10 [6 marks] "A person uses a dialogue interface to control multiple devices in their home. Evaluate the use of a dialogue interface in this context." Question paper 9626/12, June 2025, p. 9

You control a dialogue interface by speaking to it.

Full marks

Because the person controls the devices by speaking, they do not need to physically touch any switch or screen.¹ This is a real benefit for a disabled person, who can switch on lights or heating around the home without moving to them.² It also means the person can carry on with other tasks, such as cooking, while giving a command.³ However, the interface relies on sound, so loud music or a television in the room may stop it understanding the command.⁴ It may also mishear what the person says and operate the wrong device, so the person can rely on it less.⁵ Overall, a dialogue interface is very useful in a home, especially for someone with limited movement, if the rooms are quiet enough.⁶

Where the marks come from ¹ "No need to physically interact (may exemplify) (1st)". ² "More accessible to disabled users (may refer to device itself or devices around the home) (1)". ³ "Can undertake other tasks at the same time (1)". ⁴ "External noise may result in the dialogue interface not understanding the command (1)". ⁵ "Dialogue interface may misinterpret what is said (1)". ⁶ "One mark for a valid conclusion (1)". (Mark scheme 9626/12, June 2025, p. 9)

Weak answer, about 2 of 6

Dialogue interfaces are good for disabled people. You control things with your voice. It is hands-free and easy to use.

Every point is positive. No point has a consequence, and there is no conclusion. The marks for drawbacks and for the judgement are out of reach.

"Candidates tended to concentrate on the positive implications, rather than both the positive and the negative, and this did restrict the overall marks scored for this question in a few cases"

(Examiner report, June 2025, Paper 9626/12, p. 8).

Worked example

Q10 [8 marks] "A company is developing a data input screen for entering customer details into a database. Evaluate the use of evolutionary prototyping for developing the data input screen."

Question paper 9626/32, November 2024, p. 11

Full marks

In evolutionary prototyping, the developers keep refining and rebuilding the data input screen until it is complete.¹ The staff who will use the screen work with every version of it during development.² As a result, the screen is more likely to have the customer fields the staff need, which matters most for a data input screen.³ The developers can also change the screen at any iteration, for example if the company decides to record extra customer details.⁴ On the other hand, the constant meetings between developers and users can be costly.⁵ This raises the overall cost of development, which is a serious drawback when the company is building only one screen.⁶ Developers may also find it hard to know when to stop tweaking the screen, so development may go on too long.⁷ Overall, evolutionary prototyping suits this screen because user feedback matters most for data entry, if the company sets a clear end point to control cost.⁸

Where the marks come from ¹"The system / screen is continually refined and rebuilt (when developed using evolutionary prototyping)". ²"The user continuously engages with the system (1)". ³"so the new system / screen is more likely to meet the user requirements (1)". ⁴"Screen / product can be changed at any stage / iteration (1)". ⁵"Continual need to consult / interact between developers / client / end-users can be costly (1)". ⁶"which raises overall cost of development (1)". ⁷"Developers may have difficulty knowing when it is necessary to stop 'tweaking' / developing the system (1)". ⁸"One mark is available for a valid conclusion / judgement (must have considered both pros and cons)". (Mark scheme 9626/32, November 2024, p. 15)

Weak answer, about 0 to 3 of 8

Evolutionary prototyping involves the users. The screen can be changed. It can be expensive. It takes a long time.

Each line is a bare statement. Nothing follows it, and the answer never reaches a conclusion.

"Weaker answers confused the methods of prototyping, made statements with no consequences, expansion, or judgments" (Examiner report, November 2024, Paper 9626/32, p. 17).

Analyse

Analyse. Cambridge's definition: "examine in detail to show meaning, identify elements and the relationship between them" Syllabus 2025 to 2027, p. 66

Break it into points and expand them

1. Make a point, then expand it into what it causes or means. Some mark schemes say a proper analysis must have expansions. Two November 2022 mark schemes cap an answer with no expansion at 6 of 8. None of the 18 Analyse mark schemes from 2022 on gives a mark for a conclusion.

"Must have expansions to be a proper analysis" (Mark scheme 9626/11, June 2022, p. 4). *"Max six marks if no attempt at expansion"* (Mark scheme 9626/13, November 2022, p. 6).

2. A short description or definition at the start can earn one or two marks. After that, a description earns marks only when you use it to explain an effect or a reason.

"Max two from: Resolution is measured/described as pixels per inch/ppi" (Mark scheme 9626/32, March 2022, p. 5).

"Descriptions of vector images were not given credit unless these descriptions were used to explain why vector graphics are, or are not, used on websites" (Examiner report, March 2025, Paper 9626/32, p. 12).

3. Listing features gets few marks. Expand each one.

"Responses often consisted of lists of the features of encryption" (Examiner report, June 2022, Paper 9626/11, p. 2). *"The weaker answers did not go beyond listing points so could not gain many marks"* (Examiner report, June 2023, Paper 9626/31, p. 19).

Show the links

1. Show how one thing leads to another, and then to a further effect. Examiners and mark schemes reward these chains of cause and effect.

"Good answers should have made references to the need for greater bandwidth, for example, so IT infrastructure has to be upgraded which increases costs taking money from other needs" (Examiner report, June 2023, Paper 9626/31, p. 18).

"Increased digital awareness leads to increased/better infrastructure// more investment in IT (accept personal equivalences) (1) which then leads to increased digital awareness (1)" (Mark scheme 9626/12, March 2024, p. 10).

2. Cover good and bad sides where they exist. A June 2024 examiner report says full marks were possible without both. Some mark schemes cap an answer that covers only one side or one item.

"While it was possible to gain full credit without doing so, a full analysis should have given details of both 'for' and 'against' the use of the waterfall method in the question scenario" (Examiner report, June 2024, Paper 9626/32, p. 23).

"Max 4 for all lossless or for all lossy" (Mark scheme 9626/32, November 2023, p. 5).

Reach the effect the question asks for

1. Write about the effect the question asks for. If it asks about risks to an economy or effects on an image, every point must reach that.

"It was about the risks to an economy that might occur as a result of the use of digital currencies"

(Examiner report, June 2023, Paper 9626/32, p. 22). *"references to a reduction in file size without an accompanying consequence on the image itself did not answer the question"* (Examiner report, November 2023, Paper 9626/32, p. 16).

2. Keep to the exact focus of the question. Avoid empty phrases like 'looks better'.

"Questions about the digital divide require candidates to concentrate on the specific focus, otherwise answers tend to be generic, unfocused and, usually, largely irrelevant" (Examiner report, March 2024, Paper 9626/12, p. 3). *"Vague references to 'vector graphics make the site look better' or 'look attractive to students' did not gain credit"*

3. Read the whole question. If it names some tools and asks about other ones, the named tools earn nothing.

"a significant number of a candidates included the tools mentioned in the question in their answer, which did not gain them credit" (Examiner report, June 2025, Paper 9626/32, p. 20).

Write in sentences

1. Write in full sentences. 10 of the 18 Analyse mark schemes, all from 2022 and 2023, cap bullet points. The caps are 6 of 8, 5 of 7, and 4 or 5 of 6.

"questions that ask candidates to 'analyse' or 'discuss' require detailed responses that include a proper analysis or discussion, rather than short or bullet-pointed responses" (Examiner report, March 2023, Paper 9626/12, p. 1). *"Max 4 marks if bullets/list of points/characteristics"* (Mark scheme 9626/31, June 2023, p. 8).

Worked example

Q3 [6 marks] "Bitmap images can be compressed when stored and used on computer systems. Analyse the effects of the two main methods of image compression on a bitmap image."

Question paper 9626/32, November 2023, p. 4

Full marks

The two main methods of image compression are lossless compression and lossy compression.¹ Lossless compression stores all the image information exactly, because it removes no image information.² As a result, the computer can recreate the image perfectly, with no difference from the original.³ Lossy compression loses information during compression, and you cannot get that information back.⁴ Because of this, the image loses quality and may look blurred or blocky.⁵ The quality gets worse each time you compress and save the image again, so an image that someone edits many times keeps losing detail.⁶

Where the marks come from ¹ "(The two methods are) lossless compression and lossy compression". ² "All image information is stored exactly/no image information is removed". ³ "Image is/can be recreated perfectly/with no difference from the original". ⁴ "Information is lost during compression and is not/cannot be replaced". ⁵ "Images are degraded/reduced in quality/blurred/pixelated/blocky". ⁶ "Image quality is further degraded/reduced with repeated compression/decompression/saving and opening (1st)". (Mark scheme 9626/32, November 2023, p. 5)

Weak answer, about 0 or 1 of 6

Compression makes the file size smaller, so the image takes up less storage space and is quicker to send by email. Lossy makes the file smaller than lossless.

It talks about file size and sending time. The question asked about the effect on the image.

"references to a reduction in file size without an accompanying consequence on the image itself did not answer the question" (Examiner report, November 2023, Paper 9626/32, p. 16).

Justify

Justify. Cambridge's definition: "support a case with evidence/argument" Syllabus 2025 to 2027, p. 66

Argue one side only

1. Argue only for the choice in the question. Arguments against it earn nothing. The June 2025 examiner report says no Justify question needs drawbacks. Answers that gave arguments for and against did not meet the demand of the question.
"Drawbacks of using mail merge were not required when answering this question, nor are they required for any question that uses the command word 'justify'" (Examiner report, June 2025, Paper 9626/33, p. 25). "Most responses also gave arguments against the decision. These answers did not meet the demand of the question and were not worthy of credit" (Examiner report, November 2023, Paper 9626/13, p. 10).
2. You may give a drawback of an alternative, because that still supports the choice. For example, direct data lets you control the sample size, and indirect data does not.
"drawbacks do not gain credit unless they are of alternatives" (Examiner report, June 2024, Paper 9626/31, p. 20). "You are in control of the sample size//can be as big/small as needed (1st) – whereas Isabella has no control over the amount of data collected from an indirect source (1)" (Mark scheme 9626/11, June 2023, p. 10).
3. Turn each feature into a reason for using it. Examiners say many students listed features or gave descriptions, and few used them to justify the choice.
"Most candidates could list the features of a VPN, but few used these to justify its use" (Examiner report, June 2025, Paper 9626/33, p. 26). "few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions" (Examiner report, November 2025, Paper 9626/11, p. 3).

Reasons carry the marks

1. Most marks go to the reasons. 9 of the 19 Justify mark schemes from 2022 on give a first mark (1st) for a benefit. A second mark goes to the reason or result that follows from it.
"It is the reasons that gain most credit in questions that ask for 'justify'" (Examiner report, November 2024, Paper 9626/33, p. 19). "Give a positive impact (1st) • Give an extension that is a positive implication on a customer or business of the identified justification. (1)" (Mark scheme 9626/13, November 2024, p. 8).
2. Some mark schemes give one mark for a definition of the thing you justify. One clear sentence is enough.
"Any correct definition of direct data sources, such as:" (Mark scheme 9626/11, June 2023, p. 10). "Definition of encryption (1)" (Mark scheme 9626/12, November 2025, p. 8).
3. The same benefit in different words earns no extra marks. Give a range of different reasons.
"Whilst candidates were aware of some advantages, these were often narrow and repeated from different angles in an apparent attempt to achieve more marks" (Examiner report, November 2024, Paper 9626/13, p. 9). "To justify the use of an interpreter, candidates had to have a range of benefits of the use of an interpreter" (Examiner report, June 2025, Paper 9626/13, p. 11).
4. Vague reasons lose marks. Be technically precise about why the choice helps.
"some of the reasons given in support were either vague or incorrect" (Examiner report, November 2025, Paper 9626/12, p. 7). "In some cases, there was a lack of technical understanding, whilst in others, answers were vague and lacked clarity" (Examiner report, June 2025, Paper 9626/13, p. 11).
5. Write in full sentences. 4 of the 19 Justify mark schemes, all 8-mark questions from 2023, cap a bulleted list at 6 marks.
"Max 6 marks for just bullet points" (Mark scheme 9626/13, November 2023, p. 9). "Candidates should be discouraged

from writing bulleted, short statements in their responses but instead be encouraged to write answers in full sentences" (Examiner report, March 2024, Paper 9626/32, p. 7).

Tie it to the scenario

1. Link every reason to the scenario. Some mark schemes give full marks only when you join the concept to the context. A check or feature that does not fit the given data earns nothing.

"The mark scheme was structured so that full marks could only be achieved where the concept and the context were combined" (Examiner report, November 2025, Paper 9626/12, p. 7). *"a student number is unlikely to be validated with a range check, as candidates were not given a range"* (Examiner report, March 2025, Paper 9626/12, p. 3).

2. If the question names two things to justify, cover both. One March 2025 mark scheme capped answers that dealt with only one of them at 6 of 8.

"Max six if only one of student number or examination score used" (Mark scheme 9626/12, March 2025, p. 7).

Worked example

Q9 [4 marks] "A programmer is developing a new piece of software. The programmer has decided to write code that needs to be interpreted so that it will run. Justify the programmer's decision."

Question paper 9626/11, November 2025, p. 7

Full marks

With an interpreter, the programmer can write the code in a high-level language, which is easier to write and read.¹ This means the programmer does not need to code in machine code.² The same program code will run on different computers and operating systems, as long as each has an interpreter.³ The interpreter reports mistakes as soon as it finds them, so the programmer can debug the new software quickly while developing it.⁴

Where the marks come from ¹"Programmer can write in high level language (1)". ²"Programmer does not need to code in machine code (1)". ³"The program code will run on different computers/operating systems (1)". ⁴"An interpreter will highlight mistakes as they are found (1)". (Mark scheme 9626/11, November 2025, p. 8)

Weak answer, about 1 of 4

Interpreted code is easier to debug. However, it runs more slowly than compiled code and the user needs the source code, so a compiler would be better.

Only the first sentence supports the decision. The rest argues against it, which earns nothing for Justify.

"few candidates attempted to fully justify, with some candidates giving reasons against the decision, and others giving descriptions" (Examiner report, November 2025, Paper 9626/11, p. 3).

Worked example

Q10 [8 marks] "The manager of the payroll department within a company has commissioned a computer programmer to write software to run the payroll. Justify this decision to use custom written software." Question paper 9626/13, November 2023, p. 11

Full marks

The programmer creates custom written software for this exact purpose: running this company's payroll.¹ So the payroll staff can use it without adapting it first.² The programmer can make sure the software has all the features the manager requires, such as the company's own pay rules.³ It will not include unnecessary features, so payroll staff are not distracted by bloatware.⁴ The programmer can change any settings built into the software.⁵ This matters if the software does not meet the manager's requirements at first.⁶ The company will also own the copyright of the payroll software.⁷ This means it could sell the software to other businesses and earn money from it.⁸

Where the marks come from ¹"(This) custom written software is created for a defined/this purpose/running a payroll". ²"Does not need to be adapted for use (1)". ³"Software will have/Programmer can ensure the software will have all the features that are required (1)". ⁴"Custom written software will not have any unnecessary/distracting features/bloat ware (1)". ⁵"(Any) settings built into the software can be changed (1st)". ⁶"if the software doesn't initially meet the manager's requirements (1)". ⁷"The company will own the copyright of the custom written software (1st)". ⁸"so can sell it to others (1)". (Mark scheme 9626/13, November 2023, p. 9)

Weak answer, about 2 of 8

Custom software is made for the company, so it meets their needs. However, it is expensive and takes a long time to write, and it may have bugs because few people have tested it. Off-the-shelf software is cheaper and ready straight away.

The answer backs the decision once, in its first sentence. After that, the drawbacks of custom software and the benefits of off-the-shelf software argue against the manager's choice. They earn nothing.

"Most responses also gave arguments against the decision. These answers did not meet the demand of the question and were not worthy of credit" (Examiner report, November 2023, Paper 9626/13, p. 10).

Compare and Contrast

Compare. Cambridge's definition: "identify/comment on similarities and/or differences"

Syllabus 2025 to 2027, p. 66

Contrast. Cambridge's definition: "identify/comment on differences" Syllabus 2025 to 2027, p. 66

Which sides to give

1. Mark schemes for Compare list similarities and differences, with one mark for each valid point. Some mark schemes need at least one similarity and one difference for full marks. Two examples are March 2023 Paper 32 and June 2022 Paper 12.

"Must be 1 of each for full marks" (Mark scheme 9626/32, March 2023, p. 5). "Must have at least one similarity and one

difference to gain full marks" (Mark scheme 9626/12, June 2022, p. 4).

2. Some mark schemes cap each side. A 5-mark question can allow at most 4 marks for similarities and 4 for differences. An answer with one side only cannot reach full marks.

"Five marks available: Similarities Max 4" (Mark scheme 9626/11, November 2023, p. 5). *"Five from: Max 4 from: Similarities"* (Mark scheme 9626/32, November 2025, p. 6).

3. Give similarities and differences in a Compare answer, even when the question does not ask for both. The examiner reports differ here. The March 2023 report advised both sides in every case. The November 2022 report gave full marks for one side on one question. An answer with both sides is safe under either view.

"even if it does not, candidates are advised to give both to ensure access to the full mark range"

(Examiner report, March 2023, Paper 9626/32, p. 8). *"However, in this case, full marks could have been achieved with only similarities or only differences in the answers"* (Examiner report, November 2022, Paper 9626/32, p. 18).

4. Students often leave out the similarities. The November 2025 examiner report (Paper 11) said few, if any, candidates gave similarities. The June 2025 report (Paper 32) said candidates often omitted them.

"Few, if any, candidates identified similarities between the two methods"

(Examiner report, November 2025, Paper 9626/11, p. 2). *"the similarities between the two were often omitted"*

(Examiner report, June 2025, Paper 9626/32, p. 22).

5. Contrast means differences only. Similarities in a Contrast answer get no marks and use up your time.

"Any references to similarities were ignored and not given credit because these did not answer the question"

(Examiner report, June 2023, Paper 9626/32, p. 21). *"Command word: Contrast: identify/comment on differences"*

(Mark scheme 9626/32, March 2022, p. 3).

Write each point as a comparison

1. Make each point say something about both things. 'A LAN covers a small area' gets no mark. 'A LAN covers a smaller area than a WAN' gets a mark. The June 2025 report on a Contrast question said you need not write out both sides every time. The point must still read as a comparison.

"For example, 'a LAN covers a small geographical area' on its own is a true fact but does not answer the question.

However, 'a LAN covers a smaller geographical area than a WAN' answers the question and would gain credit"

(Examiner report, March 2022, Paper 9626/32, p. 6). *"whilst candidates do have to show some awareness that they are comparing, there is no requirement for candidates to present both sides of the argument"*

(Examiner report, June 2025, Paper 9626/12, p. 6).

2. Write each difference as one sentence about both things, joined by 'whereas'. Do not write one paragraph about A and then another paragraph about B. Examiners gave no marks when they could not see the comparison.

"differences are best presented together, perhaps joined by the use of 'whereas'. In some cases, candidates wrote two separate descriptions one of phishing and the other of vishing" (Examiner report, November 2022, Paper 9626/12, p. 3).

"this clarity was sometimes lacking and so marks were not awarded"

(Examiner report, November 2022, Paper 9626/12, p. 3).

3. Do not answer with a bullet list or a table. Write full sentences. One examiner report says lists of points, bulleted or in tables, are unlikely to get full marks for a comparison.

"candidates who provide lists of points, bulleted or in tables, are unlikely to gain full marks because the question asks for comparisons which are best written in free response" (Examiner report, June 2023, Paper 9626/32, p. 21). *"Weaker answers just stated facts, sometimes as unconnected bullets, or described uses of LANs/WANs"*

Compare the two named things

1. Do not repeat the question as a similarity. If the question says both devices exchange data, 'both exchange data' gets no mark. The examiners also called very general similarities weak, such as 'both are plastic cards'.

"answers that repeat the question, in this case stating that both exchange data in networks, will not be given credit"

(Examiner report, June 2023, Paper 9626/32, p. 21). *"Weaker answers referred to both being plastic cards"*

(Examiner report, June 2023, Paper 9626/31, p. 17).

2. Compare exactly the two things named in the question. Write about both of them. Do not answer about something else that just shares a keyword.

"some candidates either only focussed on one of the two methods, or did not discuss the impact of each method on the quality of a video file" (Examiner report, November 2025, Paper 9626/12, p. 7). *"a significant number of responses referred to dynamic IP assignment using DHCP. Such references did not answer the question"*

(Examiner report, March 2023, Paper 9626/32, p. 8).

Worked example

Q3 [6 marks] "Compare the use of static and dynamic routing when sending data packets through IP networks. You must include similarities and differences." Question paper 9626/32, March 2023, p. 4

Full marks

Both static and dynamic routing are used in routers to decide the next hop that a data packet is sent to.¹ Both types of routing provide a default route for a packet if no other route can be found.² Network administrators create static routing tables, whereas algorithms create dynamic routing tables.³ Static routes are fixed, whereas dynamic routing lets packets take different routes depending on network conditions.⁴ Static routing tables stay the same when network conditions or devices change, whereas dynamic routing tables update to match the changes.⁵ Static route information does not time out, whereas dynamic routes have a time-to-live.⁶

Where the marks come from ¹ "Used in routers/routing devices to determine next node/router/device/hop to send data packet/datagram/frame". ² "Both provide default route for packet if no route can be determined". ³ "Static routing tables created by network administrators whereas dynamic routing tables are created by algorithms". ⁴ "Static routes are fixed whereas dynamic routing allows packets to take different routes depending on network conditions". ⁵ "Static routing tables not updated/amended when network conditions/device change whereas dynamic routing are updated/can adjust to changing network conditions". ⁶ "Static routing table/info does not 'time out' whereas dynamic routes can be time-limited/have time-to-live data". (Mark scheme 9626/32, March 2023, p. 5)

Weak answer, about 0 of 6

Static IP addresses are set by the network administrator and do not change. Dynamic IP addresses are given out by a DHCP server and can change each time a device connects.

It answers about IP addresses, but the question is about routing. It has no similarity. It also describes each type on its own, with no comparison.

"It was noted that a significant number of responses referred to dynamic IP assignment using DHCP. Such references did not answer the question and it can only be assumed that the candidates spotted the words 'dynamic' and 'IP' and misunderstood the question" (Examiner report, March 2023, Paper 9626/32, p. 8).

Worked example

Q1 [6 marks] "Contrast the characteristics of local area networks (LANs) and wide area networks (WANs)." Question paper 9626/32, March 2022, p. 2

Full marks

A LAN covers a smaller geographical area than a WAN.¹ A LAN has a higher data transfer speed than a WAN.² One organisation usually owns a LAN, whereas a number of organisations often own a WAN.³ LANs have fewer data transmission errors than WANs.⁴ LANs have fewer network congestion problems than WANs.⁵ A LAN uses Layer 1 or 2 devices such as switches and hubs, whereas a WAN uses Layer 3 devices such as routers.⁶

Where the marks come from ¹ "LAN covers a smaller geographical area/valid examples than WAN". ² "LAN has higher data transfer speed/greater bandwidth than WAN". ³ "LANs are (usually) owned by one organisation whereas WANs are (usually/often) owned by a number/collective ownership". ⁴ "LANs have fewer data transmission errors than WANs". ⁵ "LANs have fewer network congestion problems than WANs". ⁶ "LANs use Layer 1 or 2 devices/switches/bridges/hubs/repeaters whereas WANs use Layer 3 devices/Routers/Multi-layer Switches". (Mark scheme 9626/32, March 2022, p. 3)

Weak answer, about 0 of 6

- LAN: small area, one building.
- WAN: large area, the internet.
- Both are networks that connect computers.

Each bullet states facts about one network, so no bullet gives a difference. The last bullet is a similarity, and Contrast gives no credit for similarities.

"Weaker answers just stated facts, sometimes as unconnected bullets, or described uses of LANs/WANs. Others gave similarities. These did not gain credit because the responses did not answer the question"
(Examiner report, March 2022, Paper 9626/32, p. 6).

Identify, State, Give, Define, Name, Suggest

Identify. Cambridge's definition: "name/select/recognise" Syllabus 2025 to 2027, p. 66

State. Cambridge's definition: "express in clear terms" Syllabus 2025 to 2027, p. 66

Define. Cambridge's definition: "give precise meaning" Syllabus 2025 to 2027, p. 66

Suggest. Cambridge's definition: "apply knowledge and understanding to situations where there are a range of valid responses in order to make proposals/put forward considerations" Syllabus 2025 to 2027, p. 66

Give: not in the syllabus command-word table (page 66).

Name: not in the syllabus command-word table (page 66).

List: not in the syllabus command-word table (page 66).

How much to write

1. For Identify, State and Name, a single word or short term can be enough. Examiners described one Identify question as looking for four one-word answers. Another question wanted only the name or acronym of a protocol.

"In effect, this question was a test of memory recall, as whilst the answer did not require precise wording, it was looking

for four one-word answers" (Examiner report, June 2024, Paper 9626/11, p. 2). *"This question required only a name or acronym of a protocol that works at the internet layer"* (Examiner report, June 2025, Paper 9626/31, p. 19).

2. When the question asks you to describe, write more than one word for each answer.

"As candidates were asked to describe, rather than state, answers needed to be more than one-word answers" (Examiner report, November 2025, Paper 9626/11, p. 3).

3. Short-answer questions usually give one mark for each correct item. The mark scheme says 'One from' or 'Four from'.

"ONE from: • a cryptographic hash of the previous block • a timestamp • (all the) transaction data" (Mark scheme 9626/33, June 2025, p. 7).

4. Some questions say 'identify and describe', or ask for an item and a description. The mark scheme can then use matched pairs. You get one mark for the item and one mark for a description that goes with it.

"Max two for each matched pair" (Mark scheme 9626/12, March 2024, p. 4). *"Max FOUR marks, TWO per paired answer"* (Mark scheme 9626/12, June 2025, p. 10).

5. Give exactly the number of answers that the question asks for. Examiners mark only the first answers, up to that number. Extra answers earn nothing.

"Where candidates are asked for a specific number of answers, only the first of these up to that number will be marked. Therefore, for this question, if a candidate gave more than four answers, only the first four were marked" (Examiner report, November 2023, Paper 9626/13, p. 8).

Use the exact term

1. Make a definition precise. Examiners praised short, exact definitions such as 'a sequence of characters'. They called nearly-right answers a common mistake. One example is saying that encrypted data is unreadable. In fact the data is readable, but its meaning is hidden. Learn the exact meaning of each syllabus term.

"Many candidates could give the precise definition of a string as a 'sequence of characters'" (Examiner report, November 2024, Paper 9626/33, p. 18). *"One common mistake is considering that encrypted data is unreadable without a key, which is one advantage of encrypting the data"* (Examiner report, June 2023, Paper 9626/11, p. 2).

2. Use the exact names from the syllabus. Made-up names such as 'updating maintenance' got no marks. Be as specific as the question asks. For example, 'numeric' was not enough when the question asked which numeric data type.

"Poorer answers incorrectly named the maintenance as e.g., 'updating maintenance' or 'replacement' maintenance. The names of the types of maintenance are given in Topic 16.8 so candidates are reminded to use these" (Examiner report, November 2023, Paper 9626/32, p. 18). *"Typical issues included stating that the field was numeric, but not stating what type of numeric field was best, or simply confusing the data types"* (Examiner report, November 2024, Paper 9626/12, p. 4).

Read the question closely

1. Do not give an item from the question as your answer. If the question already names an item or gives an example, repeating it earns nothing. A new wording of the question also earns nothing, for example 'a command line interface uses commands'.

"Candidates should not repeat the question in their answers. For example, Question 5(a) asks for one 'other item of data' but a significant number of candidates included the items given in the question stem in their answer, which did not gain them credit" (Examiner report, June 2025, Paper 9626/33, p. 24). *"For example, candidates stated that a command line interface used commands or a gesture-based interface used gestures. In cases where candidates answered in these terms, without any further expansion, no credit could be awarded"*

(Examiner report, November 2023, Paper 9626/12, p. 6).

2. Answer the actual question. If it asks you to give reasons, a description gets no marks. If it asks how something could go wrong, a definition gets no marks.

"Many candidates described tunnelling which did not answer the question. The question required candidates to give reasons why it would be used" (Examiner report, November 2022, Paper 9626/32, p. 16). *"Candidates who failed to achieve marks usually did so because they gave a definition of visual checking, rather than answer the question"* (Examiner report, March 2025, Paper 9626/12, p. 2).

3. Fit short answers to the scenario too. Examiners did not accept answers about the wrong part of the scenario.

"Where candidates did not achieve full marks, this was often because they listed issues to do with the train service itself" (Examiner report, November 2025, Paper 9626/12, p. 5). *"When faced with a question that has a clear context, learners should take this as an opportunity to apply their understanding in the particular context"* (Examiner report, June 2023, Paper 9626/12, p. 6).

Suggest needs reasons that fit

1. Suggest is rare, with 2 parts from 2022 onward. One of them has an examiner comment, on improving a flowchart. Each suggestion had to be relevant and had to fit clearly on the flowchart. A 2021 report on a Suggest question criticised answers that said what something was. Those answers gave no reasons that fitted the scenario.

"As the question asks for improvement to the flowchart, candidates had to ensure that suggested improvements were both relevant and able to be clearly placed on the flowchart" (Examiner report, November 2023, Paper 9626/13, p. 10). *"While many responses provided some detail about what incremental prototyping involved, most did not give reasons why it might be used when developing a spreadsheet for managing finances"* (Examiner report, June 2021, Paper 9626/32, p. 20).

Worked example

Q2 [4 marks] "Identify four factors which can affect the quality of information."

Question paper 9626/11, June 2024, p. 2

Full marks

Accuracy.¹ Relevance.² Age.³ Completeness.⁴

Where the marks come from ¹"Accuracy (1)". ²"Relevance (1)". ³"Age (1)". ⁴"Completeness (1)".

(Mark scheme 9626/11, June 2024, p. 3)

The mark scheme also accepts Level of detail.

Worked example

Q5(a) [1 mark] "Blockchains contain the records and the cryptographic hashes used to secure the records. State one other item of data that makes up the blockchain." Question paper 9626/33, June 2025, p. 6

Full marks

A timestamp.¹

Where the marks come from ¹"a timestamp". (Mark scheme 9626/33, June 2025, p. 7)

Weak answer, about 0 of 1

The cryptographic hashes that secure the records.

The question already gives this item. Repeating the question stem earns nothing.

"a significant number of candidates included the items given in the question stem in their answer, which did not gain them credit" (Examiner report, June 2025, Paper 9626/33, p. 24).

Worked example

Q1(a) [1 mark] "JavaScript can manipulate and store strings in variables. (a) Define the term string as used in JavaScript." Question paper 9626/33, November 2024, p. 2

Full marks

A string is a sequence of characters.¹

Where the marks come from ¹ *"(An object representing) a sequence of (any) characters"*.
(Mark scheme 9626/33, November 2024, p. 5)

Complete, Draw, Write (algorithms and flowcharts)

Write (an algorithm / pseudocode): not in the syllabus command-word table (page 66). This is the wording of syllabus topic 4.1. The ellipses are part of the syllabus keywords.

"Writing an algorithm using pseudocode to solve a given problem, including: – INPUT/READ – WRITE/PRINT – FOR... NEXT – FOR...NEXT...STEP – IF...ELSE...ENDIF – WHILE...ENDWHILE – REPEAT...UNTIL – CASE...ENDCASE"
(Syllabus 2025 to 2027, p. 21).

Complete (an algorithm or flowchart): not in the syllabus command-word table (page 66). This is the wording of syllabus topic 4.1.

"Writing and editing a given algorithm that demonstrates a decision-making process, including: – conditional branching – looping – nested loops – procedures/subroutines" (Syllabus 2025 to 2027, p. 21).

Draw (a flowchart): not in the syllabus command-word table (page 66). Examiners expect you to use this symbol list from syllabus page 61.

"Flowchart symbols Element Symbol Input/output Decision Terminator (Start/Stop) Process box Subroutine Connector A Flow line" (Syllabus 2025 to 2027, p. 61).

One mark per feature

1. Examiners give one mark for each feature the answer needs, such as an IF...ENDIF or outputs that match the conditions. Most of these marks stand alone, and a partly wrong answer still scores. A few marks depend on other marks. Always attempt the question. The examiner reports say the easier marks are generally accessible.

"Use of one IF...ENDIF (1) • All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)"
(Mark scheme 9626/13, June 2025, p. 7). *"While questions about flowcharts and pseudocode often test a range of skills, easier marks are generally accessible, making it disappointing to see some candidates choose not to attempt this"*

Write pseudocode the examiners accept

1. Use the structures and keywords from the syllabus (IF...ELSE...ENDIF, WHILE...ENDWHILE, REPEAT...UNTIL, CASE...ENDCASE). If the question names a structure, use it. Students who mixed CASE with IF statements lost marks.

"some candidates did not use the coding structures and terms given for this qualification"

(Examiner report, November 2023, Paper 9626/11, p. 1). *"Most candidates tried to combine the structure with IF statements, thereby highlighting this lack of understanding"* (Examiner report, June 2024, Paper 9626/11, p. 3).

2. Write the answer as pseudocode. Students who wrote prose lost marks. Cambridge publishes a pseudocode guide that shows the expected layout.

"proved challenging to candidates who wrote their answer in prose rather than using algorithmic constructs"

(Examiner report, June 2022, Paper 9626/11, p. 2). *"Centres are reminded that CIE has produced a pseudocode guide for teachers and candidates working on this course"* (Examiner report, June 2025, Paper 9626/13, p. 11).

3. Initialise counters and totals before the loop starts. In June 2025 Paper 13 Q7, initialisation inside the loop was the most common mistake among otherwise strong candidates. In Q13 of the same paper, a missing initialisation was the usual reason for losing marks.

"a sizeable minority placed the initialisation of the count within the loop"

(Examiner report, June 2025, Paper 9626/13, p. 11). *"Where candidates did not achieve full marks, this was generally because they did not initialise the variable Sum"* (Examiner report, June 2025, Paper 9626/13, p. 12).

4. Put the counter increment inside the loop. An increment that sits only inside an IF within the loop does not get the mark. Make the loop stop at the right value.

"Correct Increment of Count (1) • Increment of Count is within the loop (1) • UNTIL Count = 20 (1) • UNTIL in correct position (1)" (Mark scheme 9626/13, June 2025, p. 7). *"count ← count + 1 (if within while statement but not if within IF statement (unless in both))"* (Mark scheme 9626/12, November 2023, p. 8).

5. Small syntax errors cost marks. Write each comparison in full (IF X > Y AND X < Z, not IF X > Y and < Z). Close every IF with ENDIF. Input each variable separately ('input X and Y' does not work). Put text in speech marks, with commas between text and variables.

"treating the test criteria as a string and using the structure IF X>Y and <Z, instead of IF X>Y AND X<Z"

(Examiner report, June 2025, Paper 9626/13, p. 11). *"Using an input statement 'input X and Y'. This would not correctly input the values for the two variables. • Not indenting within the IF statement and not closing the IF statement. • Not using speech marks and commas within the outputs"* (Examiner report, November 2023, Paper 9626/13, p. 10).

6. Indent the lines inside IF, ELSE and loops. Some mark schemes give a mark just for correct indentation.

"IF and ENDIF indented correctly (1)" (Mark scheme 9626/13, November 2023, p. 8). *"Where marks were awarded, these tended to be for structural awareness, such as correct use of indentation"*

(Examiner report, June 2024, Paper 9626/11, p. 4).

7. Write variable names of two or more words with no spaces. Examiners do not take off marks for a slightly wrong variable name if it is clear which variable you mean.

"It should be noted that where variable names are made up of two or more words, there should be no gap between these words" (Examiner report, November 2022, Paper 9626/12, p. 5).

"(Ignore errors with variable names, as long as it is clear which variable is being referred to)" (Mark scheme 9626/12, November 2022, p. 8).

Draw flowcharts with the syllabus symbols

1. Use the symbols shown in the syllabus: parallelograms for input and output, rectangles for processes, and terminators for Start and Stop. Do not use other versions of the symbols.

"While it is accepted that there are variations on the use of symbols and their naming in system flowcharts, it is expected that for the purposes of this syllabus that the symbols shown in the syllabus be used"

(Examiner report, November 2024, Paper 9626/33, p. 20). "Candidates need to use the symbols provided by the board and should have practiced using them correctly" (Examiner report, March 2025, Paper 9626/12, p. 4).

2. Do not mix up input and process boxes. Mark schemes give marks for correct shapes and for arrowheads on flow lines.

"In a small number of cases, candidates confused input symbols with process symbols"

(Examiner report, November 2024, Paper 9626/12, p. 4). "All lines have arrow heads with direction of flow 1 mark 9 All correct symbols/shapes 1 mark" (Mark scheme 9626/13, June 2024, p. 7).

3. Give every decision box two exits labelled Yes and No (True and False are accepted). Write the condition as a maths expression. The examiners did not accept words such as 'is x between 5 and 10'.

"Candidates are reminded that all decision boxes must have a Yes and a No output (True and False are acceptable) and that they should use mathematical expression rather than text in their boxes. For example, the term 'is x between 5 and 10' is not acceptable" (Examiner report, June 2023, Paper 9626/11, p. 4). "each decision has two outcomes. These outcomes may be yes/no, or true/false, but they must be shown and labelled"

(Examiner report, March 2025, Paper 9626/12, p. 4).

Follow every instruction

1. Do exactly what the question asks. If it says 'redraw', draw a new flowchart and leave the printed one alone. If it asks for a flowchart, do not write pseudocode.

"candidates were clearly asked to redraw the flowchart. Candidates who amended the original flowchart and did not provide a new flowchart were therefore not answering the question" (Examiner report, June 2025, Paper 9626/13, p. 12).

"In a very small number of cases, candidates gave a form of pseudocode, instead of a flowchart for the answer" (Examiner report, November 2024, Paper 9626/12, p. 4).

2. Include every feature the question lists, such as a calculation and an output.

"As a general comment on questions such as these, most candidates did not include all features identified in the question" (Examiner report, November 2023, Paper 9626/11, p. 3).

3. When you suggest or describe a change to a flowchart or pseudocode, say exactly where it goes. For example, say which line you remove and what replaces it. Write the answer in the answer space, not on the printed code.

"candidates need to be extremely clear about which line is being removed, and how it is to be replaced"

(Examiner report, June 2025, Paper 9626/11, p. 3). "centres are advised that candidates should answer the question in the space provided in the question paper, rather than writing answers on the pseudocode included in the question paper" (Examiner report, November 2025, Paper 9626/12, p. 6).

Worked example

Q7 [8 marks] "Write a pseudocode algorithm using REPEAT...UNTIL which allows the teacher to input 20 test marks and outputs the correct message for each inputted mark."

Question paper 9626/13, June 2025, p. 6

Marks above 70 get a Distinction. Marks from 55 to 70 get a Merit, and marks from 40 to 54 get a Pass. Marks below 40 get the message 'You have not reached the Pass mark'.

Full marks

```
Count ← 0      1
REPEAT         2
    INPUT Mark  3
    IF Mark > 70 THEN
        PRINT "You have achieved a Distinction"  4
```

```

ELSE
    IF Mark > 54 THEN
        PRINT "You have achieved a Merit" 5
    ELSE
        IF Mark > 39 THEN
            PRINT "You have achieved a Pass"
        ELSE
            PRINT "You have not reached the Pass mark" 6
        ENDIF
    ENDIF
ENDIF 7
Count ← Count + 1 8
UNTIL Count = 20 9

```

Where the marks come from ¹ "Initialise Count before Mark variable is input (1)". ² "REPEAT in correct logical location (1)". ³ "Input Mark (1)". ⁴ "Correct Distinction grade criterion/correct fail criterion (1)". ⁵ "Correct Merit grade criterion/correct Pass grade criterion (1)". ⁶ "All OUTPUTs correctly match the criteria (NOT awardable if BP4 or BP5 not awarded) (1)". ⁷ "Use of one IF...ENDIF (1)". ⁸ "Correct Increment of Count (1) • Increment of Count is within the loop (1)". ⁹ "UNTIL Count = 20 (1) • UNTIL in correct position (1)". (Mark scheme 9626/13, June 2025, p. 7)

Weak answer, about 5 of 8

```

REPEAT Count ← 0 INPUT Mark IF Mark >= 55 and <= 70 THEN OUTPUT "You have achieved a Merit" ... Count ← Count + 1 UNTIL Count = 20

```

The code sets Count to 0 on every pass. The loop never ends, and the answer loses the initialisation mark. The Merit test does not repeat Mark before <= 70, and the IF has no ENDIF. The answer still gets the REPEAT, INPUT, increment and UNTIL marks.

"However, a sizeable minority placed the initialisation of the count within the loop. This was the most frequently made mistakes amongst candidates who otherwise had a strong understanding of what was required"

(Examiner report, June 2025, Paper 9626/13, p. 11).

The full-mark answer nests IF...ELSE in place of AND or >=, because the syllabus lists only the operators >, < and =.

Worked example

Q4 [7 marks] "Draw a program flowchart to represent this algorithm: • Input an integer value. • Repeat this until you have entered 20 values. • At the end, output the largest value and the average of the values." Question paper 9626/12, November 2024, p. 5

Full marks

Start with a Start terminator and finish with an End (or Stop) terminator. ¹
 In a process rectangle before the loop, set COUNT and TOTAL to 0. ²
 Use parallelograms for every input of NUM and for the output of MAX and the average. ³
 Input the first NUM and set MAX ← NUM. Then, inside the loop, use process

rectangles for $TOTAL \leftarrow TOTAL + NUM$ and $COUNT \leftarrow COUNT + 1$. ⁴

Draw two decision diamonds, 'NUM > MAX?' and 'COUNT = 20?', each with labelled Yes and No exits. ⁵

From 'NUM > MAX?', Yes goes to $MAX \leftarrow NUM$ and No skips it. From 'COUNT = 20?', No goes to an input box for the next NUM and back into the loop, and Yes leaves the loop. ⁶

After the loop, calculate $AVERAGE \leftarrow TOTAL / 20$ in a process rectangle. ⁷

Where the marks come from ¹"Start and End (accept STOP) terminators (1)". ²"Two variables (e.g. MAX, COUNT, TOTAL, NUM) initialised/set to zero before loop (1)". ³"Input box and output box are both parallelograms (1)". ⁴"Sum all values input (1) 11 Set new MAX when input value > MAX in loop 12 Increment COUNT (1)". ⁵"TWO Correctly drawn decision boxes with YES/NO decisions (1)". ⁶"All YES/NO decisions (used) go to correct outcomes (i.e. all logic is correct) (1)". ⁷"Correctly calculate average (must include process/calculation) (1)".

(Mark scheme 9626/12, November 2024, p. 5)

Weak answer, about 2 of 7

A flowchart that puts 'Input number' in a rectangle, has a decision box 'Have 20 numbers been entered' with no Yes or No labels, and never sets COUNT or TOTAL to zero.

The input sits in a rectangle, which is the wrong symbol. The decision is in words and has no Yes or No labels. COUNT and TOTAL never start at zero.

"In a small number of cases, candidates confused input symbols with process symbols, but overall, performance on this question was very good" (Examiner report, November 2024, Paper 9626/12, p. 4).

Setting MAX to the first value makes the flowchart work when every number is negative, and the mark scheme also accepts MAX set to 0.

About this booklet

- Sources: the 9626 syllabus (2025 to 2027), the Cambridge Learner Guide for 9626, and Paper 1 and Paper 3 question papers, mark schemes and examiner reports, 2017 to March 2026.
- Quotations: 373, from 67 documents, each checked word for word against its page.
- Counts: Papers 1 and 3, 2022 to March 2026. A part with no command word of its own counts as none.
- Worked examples: real questions. Full-mark answers are mark scheme points written as sentences. Weak answers show mistakes the examiner reports describe. No candidate wrote them.
- Not used: Cambridge's Example Candidate Responses (School Support Hub, teacher login). Papers 2 and 4.

How much evidence there is for each word

Describe	409 parts, and 388 of them have a mark scheme. 83 of those 388 mark schemes use the (1st) label. Cambridge's mark-scheme key does not define (1st).
Explain	176 parts, and 174 of them have a mark scheme. 38 of those 174 mark schemes use the (1st) and (1) labels. The rest give one mark for each point. Only one mark scheme limits answers that are 'all only first marks'. It is Q3(c) on Paper 13, November 2023.
Discuss	80 parts. 77 have mark schemes, all from 2022 to 2025. The 3 parts from March 2026 have none. The 2025 mark schemes no longer print the bullet cap or the both-sides rule. For 2025, the advice on format comes from examiner reports.
Evaluate (and Assess)	Evaluate has 25 parts. 24 have mark schemes, all from 2022 to 2025. The conclusion mark appears only from November 2024, in 8 of 9 mark schemes. Assess is in the syllabus list, but no question part from 2017 on uses it. With no Assess mark schemes or examiner comments, the syllabus definition is the only advice for Assess.
Analyse	19 parts, 13 of them on Paper 3. 18 have mark schemes, all from 2022 to 2025. One examiner report (June 2024, Paper 32, Q8) says full marks were possible without both sides.
Justify	20 parts. 19 have mark schemes, all from 2022 to 2025. March 2026 Q10(c) has none. Examiner reports from 2022 to 2025 repeat the one-sided rule. Three sources back the exception for drawbacks of alternatives. They are one examiner report sentence (June 2024, Papers 31 and 33), one 2023 comment and one mark scheme.

Compare and Contrast	Compare: 16 question parts from 2022 to 2025, all 16 with mark schemes. Contrast: 6 parts from 2022 to 2026, 5 with mark schemes. The sixth part, March 2026 Paper 12 Q7(b), has no mark scheme in the source set. November 2024 Paper 33 Q3(b) is left out because 'contrast' there is an image term in a Describe question. The examiner reports disagree on whether Compare needs both sides when the question does not ask for both.
Identify, State, Give, Define, Name, Suggest	Parts: Identify 28, State 12, Give 11, Define 8, List 2, Suggest 2, Name 1. Parts with a mark scheme: Identify 26, State 8, Give 11, Define 6, List 2, Suggest 2, Name 0. The two List parts also ask you to describe. The one Name part (March 2026, Paper 12 Q10) says 'identify' in its wording. One Suggest question from 2022 onward has an examiner comment (Q8 on Paper 13, November 2023). The Suggest advice also uses one 2021 report. Give, Name and List are not in the syllabus command-word table. No examiner report says whether an example in a definition gains or loses marks.
Complete, Draw, Write (algorithms and flowcharts)	Write: 7 question parts from 2022 to 2025, all 7 with mark schemes. Complete: 17 parts from 2022 to 2026, 15 with mark schemes. Draw: 9 parts from 2022 to 2025, 8 with mark schemes. Most algorithm and program-flowchart questions are in Paper 1. In Paper 3, Draw and Complete also tag DFD, PERT and system-flowchart parts, and only the symbol points apply to those. Write also tags November 2025 Paper 11 and Paper 13 Q8(b), which are about audio. The June 2022 examiner report says several times that algorithms and flowcharts were a recent addition to the syllabus.